



CIA2C14 Data Science Essentials

Data Acquisition & Data Understanding

What is Raw Data

- XML
- HTML
- JSON
- TEXT
- MYSQL
- MongoDB

❑ The Data that you want

- Where is it located?
- How accessible is it?
- What is structure / format ?

❑ Location of Data

- Is it in one single specific place? (e.g. one HTML table, one file in the Web)
- Is it in one website but spread across several pages? (eg several HTML tables at different pages)
- Is it spread across several websites? (e.g. multiple pieces of information in various sites)
- Is it in one or several databases?

What is Raw Data

www.gutenberg.org/ebooks/2701.txt.utf-8

The Project Gutenberg eBook of Moby Dick; or The Whale, by Herman Melville

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Author: Herman Melville

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Language: English

*** START OF THIS PROJECT GUTENBERG EBOOK MOBY DICK; OR THE WHALE ***

Produced by Daniel Lazarus and Josesey

cars2004

File Edit View Insert Format Data Tools Help All changes saved in Drive

Model

	A	B	C	D	E	F	G
	Model	Cylinders	Horsepower	Speed	Weight	Value	Length
1	Citrus CD 1.1 Base	1124	61	158	932	1839	3096
2	Smart Fortwo Coupe	656	52	135	730	1515	2530
3	Mini 1.8 170	1920	170	218	1215	1893	3525
4	Nissan Versa 1.2 850	1242	85	134	980	1887	3719
5	Renault Clio 3.0 VR	2940	295	245	1488	1819	3812
6	Audi A3 1.8 TDI	1965	185	187	1285	1785	4303
7	Porsche 911 Carrera 4S	1380	70	162	1178	1746	4202
8	Porsche 911 Carrera 4S	3940	211	229	1640	1811	4076
9	Mercedes-Benz C 270 CDI	2980	170	230	1888	1728	4328
10	Mercedes-Benz C 270 CDI	2910	218	240	1888	1848	4841
11	Audi A8 4.2 V8	4700	287	230	1712	1818	4905
12	Audi A8 4.2 V8	4380	323	280	1879	1922	5029
13	Mercedes-Benz C 400 CDI	3950	280	230	1915	2082	5038
14	Citrus CD 1.1 Base	1957	110	185	1177	1793	3934
15	BMW 2.0 16	2045	182	230	1288	1781	4091
16	Audi TT 1.8T 180	1781	180	228	1288	1784	4041
17	Audi TT 1.8T 180	3925	480	300	1825	1923	4905

World record progression

en.wikipedia.org/wiki/World_record_progression_1500...

#	Time	Name	Nationality	Date	Meet	Location
1	22:48.4	Harry Taylor	Great Britain	Jul 25, 1908	Olympic Games	London, United Kingdom
2	22:00.0	George Hodgson	Canada	Jul 10, 1912	Olympic Games	Stockholm, Sweden
3	21:55.3	Arne Borg	Sweden	Jul 8, 1923	-	Gothenburg, Sweden
4	21:15.0	Arne Borg	Sweden	Jan 30, 1924	-	Sydney, Australia
5	21:11.4	Arne Borg	Sweden	Jul 13, 1924	-	Paris, France

Accessibility of Data

- Do you have free, direct, or immediate access to data?
- Do you need to use an API?
- Do you require username, password, authentication?
- Do you need to use a specific transfer protocol?
- Do you need to use a specific type/method of request?

Format / Structure of Data

- Is it plain text?
- Is it in tabular (spreadsheet-like) form?
- Is it in HTML? XML? JSON?
- Other formats: binary, images, maps, etc?

Data needs to be processed

- Raw Data are typically NOT ready to be analyzed
- For a variety of reasons:
 - Missing values
 - Inconsistencies in data entry
 - Different formats
- Hence some processing is needed to make it usable:

Processed data

- Data that is ready for analysis
- Processing can include merging, subsetting, transforming, etc.
- There may be standards for processing
- All steps should be recorded *some data may only need to process once!*

States of Data

Data at rest

- All data in computer storage that is not currently being accessed or transferred
- Can be structured or unstructured
- Take precautions such as data encryption, hierarchical password protection, secure server rooms to prevent threats posed by hackers

Data in motion

- Data that is moving or being transferred between locations (e.g. from cloud to hard disk) within or between computer systems.
- Ready to be updated, processed, accessed and read.
- Also prone to hacking:
 - encrypting the data before it is transmitted (while in a data at rest state)
 - encrypting the passage along which the data is sent.
- Can also be semi-structured or unstructured data.

Data in use

- Data from a data warehouse/lake
- It is typically in a **structured format**, such as in relational tables.
- Directly accessible by one or more users
- Most vulnerable to attacks
- Addition to encryption, protect the data using authentication of users at all stages, restricted access levels for different staff, etc.

Possible for data to change states!

Crux of Data Form

The tidy data

1. Each variable you measure should be in one column
2. Each different observation of that variable should be in a different row
3. There should be one table for each "kind" of variable
4. If you have multiple tables, they should include a column in the table that allows them to be linked

Some other important tips

- Include a row at the top of each file with variable names.
- Make variable names human readable AgeAtDiagnosis instead of AgeDx
- In general data should be saved in one file per table.

Make our downstream activities easier...

As a start, keep to only one table/DataFrame



Data are Different

Name	DOB	Race
A	20012001	2
B	01061999	7

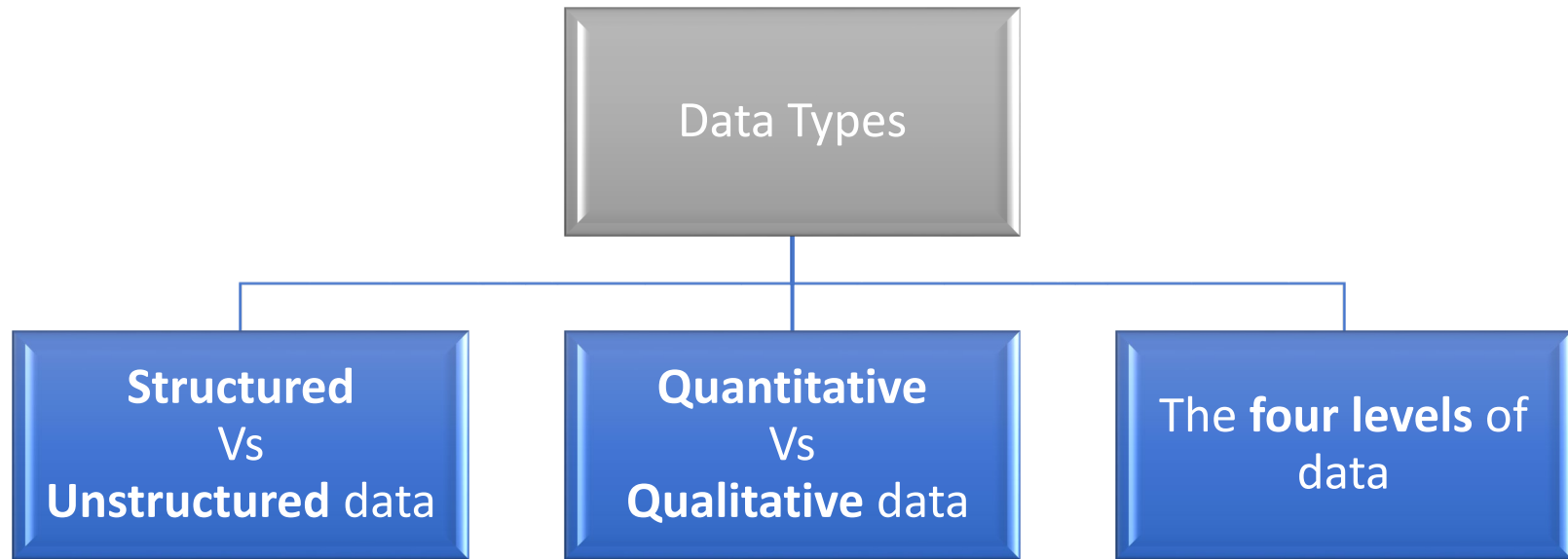
Race	Denoted by
Asian American	2
Caucasian	7

What do they mean?



- When given any dataset, it is tempting to jump right into exploring, applying statistical models, and researching how to apply machine learning to get fast outcomes.
- Resources will be wasted if we spent time trying to optimise an algorithm which does not even fit the data in the first place.
- Key: **Data Understanding**
- Different statistical/mathematical models are more effective on certain data types.

Understand your Data



Structured Vs Unstructured

Structured (organized)

- Think: Tabular form (with rows and columns)
- Rows: Observations/ Instances
- Columns: Variables/ Attributes

Unstructured (unorganized)

- This data exists as a free entity
- Does not follow any standard organization hierarchy.

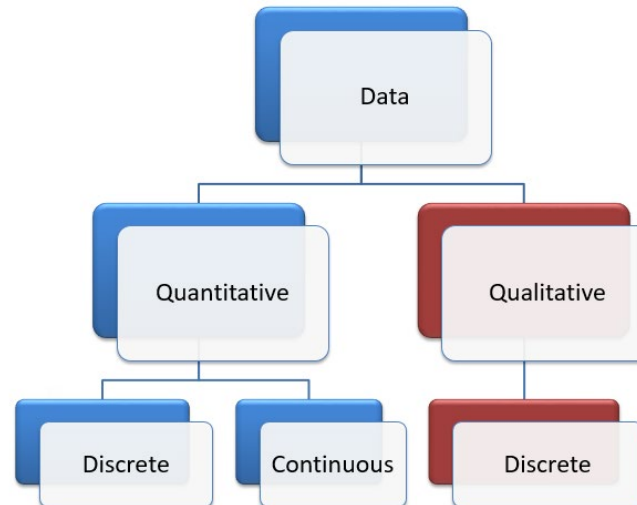
Quantitative Vs Qualitative

Quantitative/ Numerical

- Described using numbers, and basic mathematics
- Rule of Thumb: The summed value is meaningful

Qualitative/ Categorical

- Cannot be described using numbers and basic mathematics.
- Generally thought of as being described using "natural" categories and language.



Test your understanding (I)

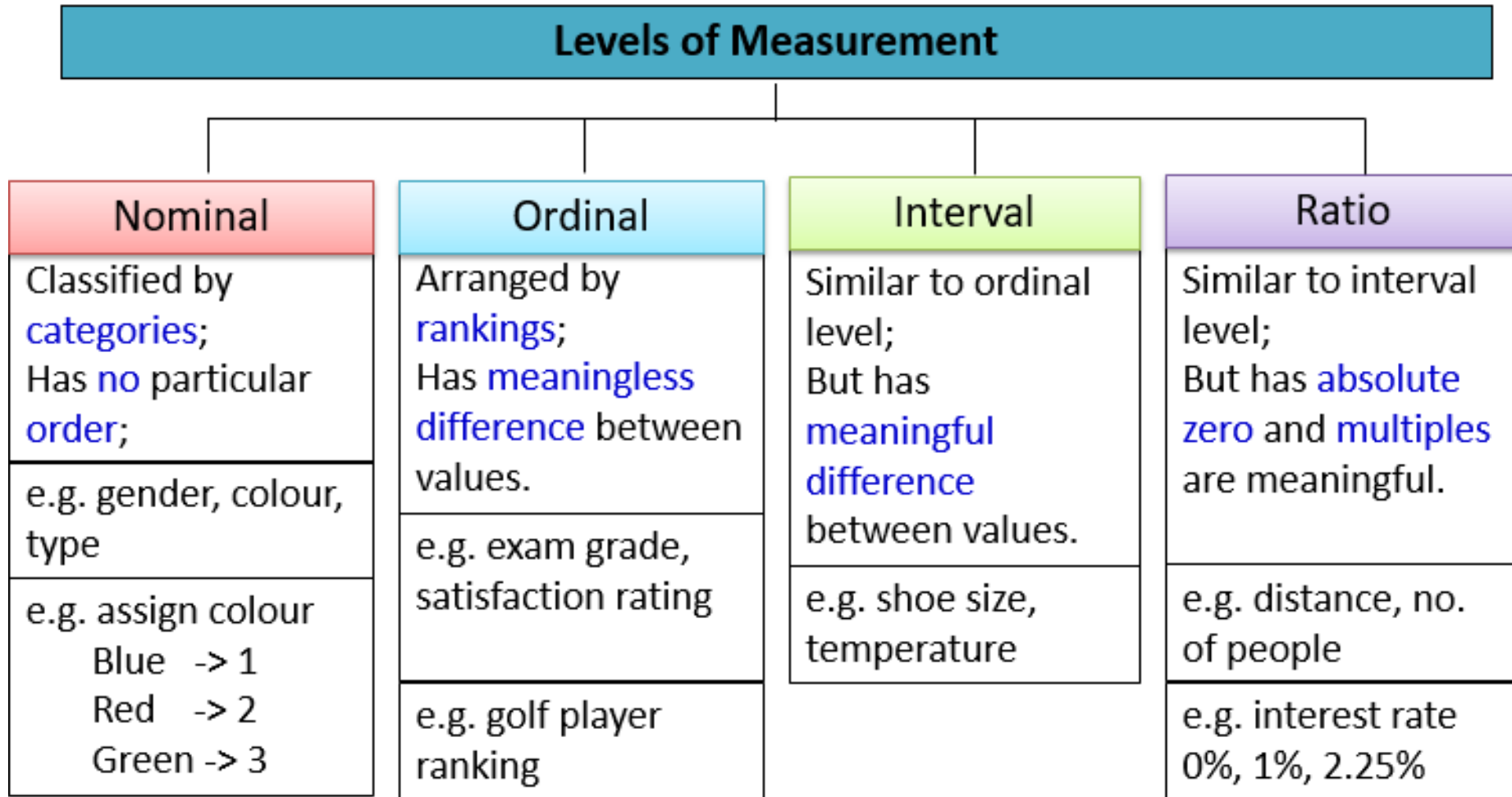
quantitative or qualitative?

Data	Type
Name of coffee shop	
Revenue	
Zip code	
Average monthly customers	
Country of coffee origin	

Test your understanding (II)

	Qualitative (& Discrete)	Quantitative	
		Discrete	Continuous
No. of durians sold			
Types of durians			
Time for transporting durians			
Amount of rainfall in June			
Types of places that sell durians			
Monthly salary range of durian sellers			
Daily temperature in June			

4 Levels of Measurement



Test your understanding (III)

	Nominal	Ordinal	Interval	Ratio
Age of students				
Font size				
Type of housing				
Outside temperature				
Number of buses				
Ranking of schools				
Classification of living things				
Speed of the car				

What's next

Whenever you are faced with a new dataset, the first three questions you should ask about it are the following:

- Is the data organized or unorganized?
 - For example, does our data exist in a nice, clean row/column structure?
- Is each column quantitative or qualitative?
 - For example, are the values numbers, strings, or do they represent quantities?
- At what level of data is each column?
 - For example, are the values at the nominal, ordinal, interval, or ratio level?

- ❑ The answers to these questions will dictate the next steps of your analysis:
 - > The types of graphs for visualization
 - > The types of predictive models
- ❑ Sometimes we will have to convert from one level to another.

**WE 
DATA
SCIENCE**